11. Sequences and Series

11.10. Taylor and Maclaurin Series

11.10 Learning Objectives

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11.10.1. Maclaurin Series of Common Functions

You should be able to recognize (or re-derive) the following functions and their series.

Function	Series	Interval of Convergence
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$(-\infty, \infty)$
b^x	$\sum_{n=0}^{\infty} \frac{b(\ln(b))^n}{n!} x^n$	$(-\infty, \infty)$
$\cos(x)$	$\sum_{*n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$(-\infty, \infty)$

Exercise 11.10.1

Derive the above Maclaurin series expansions and intervals of convergence for the above series.

11.10a Section Summary: